

Starting from the Ornstein-Zernike (OZ) integral equation

$$h_{ij}(r) = c_{ij}(r) + \sum_{k=1}^n \rho_k \int h_{ik}(t) c_{kj}(|\vec{r}-\vec{t}|) dV, \quad (1)$$

we multiply both sides by $\sqrt{\rho_i \rho_j}$, obtaining

$$H(r) = C(r) + \int H(t) C(|\vec{r}-\vec{t}|) dV \quad (2)$$

where $H(r)$ and $C(r)$ are $n \times n$ matrices with elements $\sqrt{\rho_i \rho_j} h_{ij}(r)$ and $\sqrt{\rho_i \rho_j} c_{ij}(r)$, respectively.

Now, by taking the Fourier transform of the matrix form of the OZ equation, we get

$$\hat{H}(k) = \hat{C}(k) + \hat{H}(k) \hat{C}(k) \quad (3)$$

which is equivalent to

$$\left[I - \hat{C}(k) \right] \left[\hat{H}(k) + I \right] = I \quad (4)$$

where I is the identity matrix while $\hat{H}(k)$ and $\hat{C}(k)$ are $n \times n$ matrices with elements of the form

$$\hat{H}_{ij}(k) = 4\pi \sqrt{\rho_i \rho_j} \int_0^\infty h_{ij}(r) \frac{\sin(kr)}{kr} r^2 dr \quad (5)$$

$$\hat{C}_{ij}(k) = 4\pi \sqrt{\rho_i \rho_j} \int_0^\infty c_{ij}(r) \frac{\sin(kr)}{kr} r^2 dr \quad (6)$$

By solving Eq. 4, it is possible to calculate the partial structure factors for a system of n ionic species; nonetheless, the long-ranged Coulombic interactions may produce $\hat{C}_{ij}(k)$ noisy profiles since $c_{ij}(r)$ does not decay to zero fast enough. For that reason, we may want enclose this Coulombic interactions and deal with them separately from $c_{ij}(r)$. In order to do this, we define $F(r)$ as

$$F(r) = C(r) - \Phi(r) \quad (7)$$

where $\Phi(r)$ is a $n \times n$ matrix with elements $\phi_{ij}(r) = -\beta q_i q_j / (\epsilon r)$.

It is our intention to introduce $F(r)$ into Eq. 4 but, for that purpose, we must find a relation between $\Phi(r)$ and $Q(r)$, a function which decays fast enough to zero, with elements of the form $q_{ij}(r) = -\beta q_i q_j e^{-\kappa r} / (\epsilon r)$. This relation can be found when using the Debye-Huckel closure ($c_{ij} = \phi_{ij}(r)$) in Eq. 1, which results in $h_{ij}(r) = q_{ij}(r)$.

In this way, we can establish the following relation

$$I - \hat{\Phi}(k) = \left[I + \hat{Q}(k) \right]^{-1} \quad (8)$$

where $\hat{Q}(k)$ is a $n \times n$ matrix with elements

$$\hat{Q}_{ij}(k) = 4\pi \sqrt{\rho_i \rho_j} \int_0^\infty q_{ij}(r) \frac{\sin(kr)}{kr} r^2 dr \quad (9)$$

With the above relations we are ready to introduce $F(r)$ into Eq. 4 by substituting $\hat{C}(k) = \hat{F}(k) + \hat{\Phi}(k)$, obtaining

$$\left[I - (\hat{F} + \hat{\Phi}) \right] \left[\hat{H} + I \right] = I \quad (10)$$

reordering some terms

$$\left[(I - \hat{\Phi}) - \hat{F} \right] \left[\hat{H} + I \right] = I \quad (11)$$

using Eq. 8

$$\left[(I + \hat{Q})^{-1} - \hat{F} \right] \left[\hat{H} + I \right] = I \quad (12)$$

distributing $\hat{H} + I$

$$\left[I + \hat{Q} \right]^{-1} \left[\hat{H} + I \right] - \hat{F} \left[\hat{H} + I \right] = I \quad (13)$$

adding $\hat{F} \left[\hat{H} + I \right]$ to both sides

$$\left[I + \hat{Q} \right]^{-1} \left[\hat{H} + I \right] = I + \hat{F} \left[\hat{H} + I \right] \quad (14)$$

distributing $\hat{F}$

$$\left[I + \hat{Q} \right]^{-1} \left[\hat{H} + I \right] = I + \hat{F} \hat{H} + \hat{F} \quad (15)$$

multiplying from the left both sides by $\left[I + \hat{Q} \right]$

$$\hat{H} + I = \left[I + \hat{Q} \right] \left[I + \hat{F} \hat{H} + \hat{F} \right] \quad (16)$$

distributing $I + \hat{Q}$

$$\hat{H} + I = I + \hat{Q} + \left[I + \hat{Q} \right] \hat{F} \hat{H} + \left[I + \hat{Q} \right] \hat{F} \quad (17)$$

$$\hat{H} = \hat{Q} + \left[I + \hat{Q} \right] \hat{F} \hat{H} + \left[I + \hat{Q} \right] \hat{F} \quad (18)$$

$$\hat{H} = \hat{Q} + \left[I + \hat{Q} \right] \hat{F} \hat{H} + \hat{F} + \hat{Q} \hat{F} \quad (19)$$

$$\hat{H} = \left(\hat{Q} + \hat{F} \right) + \left[I + \hat{Q} \right] \hat{F} \hat{H} + \hat{Q} \hat{F} \quad (20)$$

subtracting $\left[\hat{Q} + \hat{F} + \left[I + \hat{Q} \right] \hat{F} \hat{H} \right]$ from both sides

$$\hat{H} - \left(\hat{Q} + \hat{F} \right) - \left[I + \hat{Q} \right] \hat{F} \hat{H} = \hat{Q} \hat{F} \quad (21)$$

factoring $\hat{H}$

$$\left(I - \left[I + \hat{Q} \right] \hat{F} \right) \hat{H} - \left(\hat{Q} + \hat{F} \right) = \hat{Q} \hat{F} \quad (22)$$

adding $\left[I + \hat{Q} \right] \hat{F} \left[\hat{Q} + \hat{F} \right]$ to both sides

$$\left(I - \left[I + \hat{Q} \right] \hat{F} \right) \hat{H} - \left(\hat{Q} + \hat{F} \right) + \left[I + \hat{Q} \right] \hat{F} \left[\hat{Q} + \hat{F} \right] = \hat{Q} \hat{F} + \left[I + \hat{Q} \right] \hat{F} \left[\hat{Q} + \hat{F} \right] \quad (23)$$

factoring $\left[\hat{Q} + \hat{F} \right]$

$$\left(I - \left[I + \hat{Q} \right] \hat{F} \right) \hat{H} - \left(I - \left[I + \hat{Q} \right] \hat{F} \right) \left[\hat{Q} + \hat{F} \right] = \hat{Q} \hat{F} + \left[I + \hat{Q} \right] \hat{F} \left[\hat{Q} + \hat{F} \right] \quad (24)$$

factoring $\left(I - \left[I + \hat{Q} \right] \hat{F} \right)$

$$\left(I - \left[I + \hat{Q} \right] \hat{F} \right) \left(\hat{H} - \left[\hat{Q} + \hat{F} \right] \right) = \hat{Q} \hat{F} + \left[I + \hat{Q} \right] \hat{F} \left[\hat{Q} + \hat{F} \right] \quad (25)$$

multiplying from the left both sides by $\left[I - \left(I + \hat{Q} \right) \hat{F} \right]^{-1}$

$$\hat{H} - \left[\hat{Q} + \hat{F} \right] = \left[I - \left(I + \hat{Q} \right) \hat{F} \right]^{-1} \left[\hat{Q} \hat{F} + \left(I + \hat{Q} \right) \hat{F} \left(\hat{Q} + \hat{F} \right) \right] \quad (26)$$

adding $\hat{Q} + \hat{F}$ to both sides

$$\hat{H} = \hat{Q} + \hat{F} + \left[I - \left(I + \hat{Q} \right) \hat{F} \right]^{-1} \left[\hat{Q} \hat{F} + \left(I + \hat{Q} \right) \hat{F} \left(\hat{Q} + \hat{F} \right) \right] \quad (27)$$

After carrying out all this math derivation, we arrive at Eq. 27, which is equivalent to Eq. 3.